

Module:	MAT602 (First sit 2017)	Title:	Materials Selection in Design
First examiner:	James Busfield	Second examiner:	Raza Shah

Qu. No.	Worked Solution	Marks
	1. Which of the following class of materials typically has the highest specific modulus? a. Metals b. Plastics c. Ceramics d. Elastomers x 2. Which of the following materials groups contains materials that are the most difficult to recycle? a. Glass b. Thermoplastic c. Glass fibre reinforced polymer composite d. Metal alloy x 3. Fracture toughness K_{IC} is typically expressed using which of the following SI units? a. $\text{MPa m}^{1/2}$ b. $\text{MPa m}^{-1/2}$ c. $\text{kJ m}^{-1/2}$ d. kJ m^2 x 4. If the linear thermal-expansion coefficient of a thermally isotropic material is α / K^{-1} then which of the following gives an approximate value for the volumetric thermal expansion coefficient? a. α b. 2α c. 3α d. 4α x 5. The Wiedemann-Franz law is best summarised by which of the following statements? a. The elastic modulus of different materials are linearly related to their yield strength. b. The strength of different materials are linearly related to their density. c. The electrical and thermal conductivities of materials are linearly related. x d. The toughness and thermal conductivity of materials are linearly related. 6. Which of the following materials index values, where E is the elastic tensile modulus and ρ is the density, should be maximised to identify the best materials for a lightweight stiff beam? a. $E^{1/3}/\rho$ b. E/ρ c. E^2/ρ d. $E^{1/2}/\rho$ x 7. Which of the following materials as a result of its physical properties make the most suitable electrical insulators for spark plugs? a. Aluminium Nitride b. Alumina c. Silica d. Silicon Nitride x	+1 each that is correct -1/3 for each wrong answer

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	8. For a device to be useful in a sealing application it must have which of the following properties? a. A high modulus b. A high contact pressure c. A large contact area x d. Significant visco-elastic behaviour 9. Spray up is a specific form of contact moulding used in the manufacture of polymer composites. It layers up the fibres and the resin simultaneously using a hand controlled mixing spray system. Which of the following is a potential advantage of using spray up when compared to a hand lay up process? a. It produces a higher quality surface finish. b. The process is faster. x c. The curing time is reduced d. It layers up the fibres in a less random way. 10. The crystalline structures that form in metals and ceramics are not perfect, they contain defects that are known as? a. relocations b. dislocations x c. linear defects d. polar defects 11. Which of the following metals has the lowest density? a. Magnesium x b. Iron c. Copper d. Steel 12. The hardenability of a metal can be determined using which of the following tests? a. Jominy test x b. Jiminy test c. Jogerty test d. Jigerty test 13. At very large production volumes of more than 10,000,000 parts per year which of the following processing routes is the most economic for a simple polymer component such as a Lego™ toy brick? a. Contact moulding b. Transfer moulding c. Rotational moulding d. Injection moulding x 14. Which of the following statements is generally true about thermosetting polymers? a. They polymerise to form a three dimensional network. x b. They can be re-melted. c. They are commonly solid before processing and are easily melted by the action of heat.	

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	d. The chemical reaction transforming the initial liquid into the final 3-D macromolecule, takes place in the mixer prior to moulding. 15. Which of the following statements is the most accurate description of when some polymeric materials exhibit rubber-like behaviour? - The forces between neighbouring molecules are covalent. - The polymer must be highly crystalline. - The polymer should be hydrophilic. - The polymer must be above the glass transition temperature. x 16. According to a 2012 EU study what is the approximate size of the EU plastics industry - It employ 15,000,000 people and has a turnover in excess of €2,880,000,000,000 - It employ 1,500,000 people and has a turnover in excess of €288,000,000,000 x - It employ 150,000 people and has a turnover in excess of €28,800,000,000 - It employ 15,000 people and has a turnover in excess of €2,880,000,000 17. According to the most recent Eurostat survey which of the following industrial uses of ceramic materials is the largest by commercial value in Europe? - Technical ceramics - Floor and wall tiles x - Sanitary ware - Tableware 18. When choosing the most appropriate manufacturing process for an aluminium con-rod used in a modern automotive engineering application the choice of the most suitable manufacturing process is dependent upon the production volume. For the case when we anticipate manufacturing more than 100,000 parts per annum, which of the following processes is the most cost effective per part. - Sand Casting - Low pressure casting - Die casting x - Investment casting 19. When multiple requirements specify two conflicting goals for a material's selection process then it is necessary to devise a way of determining the best solution. One approach is to derive metric that can describe each goal. You can then draw a graph of one metric against the other and evaluate the performance of different materials against each other. A trade off surface appears whereby the best performing materials can be identified. This surface is known as the - Lorimer front - Pareto front x - Clarke front - Charles front 20. When multiple requirements specify two conflicting goals for a material's selection then it is necessary to devise a way of determining the best solution. One approach is to derive metric that can describe each goal. You can then draw a graph of one metric against the other and evaluate the performance of different materials against each other. In this approach a dominated solution is.	

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	a. One that is optimised by both metrics b. One that is optimised by only one metric c. One where the poor performance for one metric is dominated by an excellent performance by the other metric. d. One that is not optimal by either metric x 21. A full life cycle analysis can reveal which element of a specific product's life has the greatest environmental impact. Using the specific example of a 2 litre PET bottle used to package and transport water from France to the UK, which of the following life phases will consume the greatest amount of energy and consequently produce the greatest amount of CO₂? a. The production of the material. x b. The manufacture of the bottle. c. The transportation of the bottle. d. The usage of the bottle. 22. One of the best ways to reduce the complexity of a finite element analysis package is to exploit symmetry to reduce the size of the problem. Which of the following is a standard symmetric approximation used in practice. a. Planar symmetry b. Axial symmetry c. Cyclic symmetry d. All of the above. x 23. Which of the following is the most essential property for the matrix material when making thermoset polymer composites? a. A low viscosity x b. A high melting temperature c. A short pot life d. A high molecular weight prior to curing 24. Contact moulding is the preferred route for manufacturing some types of polymer composite components. Which of the following is a significant disadvantage with using contact moulding? a. There is only one good surface finish. x b. The mould is expensive. c. Large complex shapes may not be easily moulded d. Expensive and specialised equipment is required. 25. Which of the following materials is recommended by Schlumberger as the most suitable for cutting through rock under subsea conditions. a. 4340 Steel. b. Tungsten carbide. c. Silicon carbide. d. Polycrystalline diamond. x	

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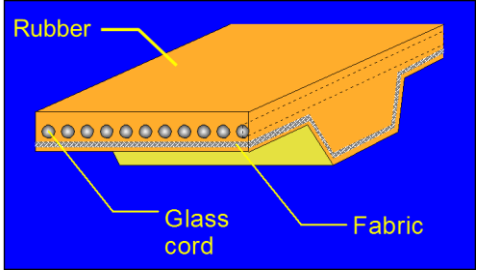
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Section B1	As a manufacturer of bicycle components, you are considering entering the market for front forks for a mountain bike. Your brand niche is for amateur enthusiast who is prepared to pay up to £400. Your business plan has considered an annual sales volume of about 10,000 units a year. a. Sketch a suitable design for the front forks for a mountain bike that is to be commercially viable at a retail price below £400 and with sales volumes of about 10,000 units a year. Your target market is an amateur enthusiast. b. What are the commercial considerations for this product? c. For each of the components identified in your earlier sketch, suggest with reasons what you would make the components from? d. What joining methods do you envisage using to make the final assembly? a. The diagram should explain how the fork attaches to the frame and the handle bars. The following parts should be labelled from the top, steerer (or headset) tube, : Crown, composite tubes, drop outs b. This is a premium sector for the serious enthusiast or even semi-professional - therefore the price is at the upper end of the range and is not particularly sensitive. The materials should be perceived as being high tech with good aesthetics and image. The volumes are low, so the forks are to be hand built and assembled. Due to the low production volumes the tooling costs should be minimised. However jigs will be required to ensure that the adhesive joints are held whilst they set overnight. An oven is also required to heat the crown for the shrink fitting. One person can make ~50 parts per day in a small area of say about 12m². The amount of stock carried is reduced using just in time delivery and despatch. The forks should be light, with high performance and appear high tech, for this the customer pays extra. Ideally you should have as many pre finished surfaces as possible and aim to damage these as little as possible during build. The epoxy cure can be left overnight to complete to avoid using an oven. c. Steerer tube is made out of steel (could be aluminium). The high strength requirement as well as a good bearing surface against the frame which requires a hard finish make it a better option than a composite part. Made using an electrically welded wrapped tube. Steel being strong and stiff can be of smaller wall thickness than other metals. The crown is made from a hollow aluminium die (could be sand) casting. The outer surface can be anodised and the bores are accurately finished and anodised to meet the other mating surfaces. The down tubes are carbon fiber epoxy resin composite tubes that are cut from filament wound long tubes. These have excellent specific strength and stiffness. Pultrusion can also be used with a twist. (answers that say carbon or just carbon fibre don't get full credit as it is clearly a composite) Magnesium drop outs are used for the excellent specific strength, low density and because of the high tech image. d. The entire assembly is bonded together. An one part anaerobic adhesive (Henkel Loctite 640) is used to join the aluminium crown to the steerer tube. This is a tight interference fit so that the crown must be first heated to expand it and then the adhesive is applied to the tube. It is quickly inserted using an accurate jig. The curing at the elevated temperature is of the order of seconds. A two part epoxy based system (Loctite 3425) is used to join the drop outs and the crown to the composite tubes. For the crown to the tubes the adhesive is mixed in a mixing head and then it is injected through a small orifice into the pre-assembled and accurately jigged frame. This allows the glue to fully fill out the joint area. Once it is seen to leak out the other side the joint is seen to be filled. For the joint between the drop outs to the tube: the different components of the adhesive can be added to each and then they are push fit together. Again excess adhesive is carefully wiped away.	5 marks 5marks 10 marks 5 marks



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B2	Fracture fixation devices are used for a wide range of different applications in the body to repair bones. One specific fracture fixation device is known as a trochanteric femoral nail. a) What is a trochanteric femoral nail and when and how is it used? [6 marks] b) What are the key features and benefits of this type of fracture fixation device when compared to other similar types of fracture fixation device? [7 marks] c) The system contains five different components, identify each and select the most suitable material and manufacturing route that should be used to make each? [12 marks]  a. The trochanteric femoral nail is a short (~200mm long) intramedullary (up the middle) nail for treatment of fractures in the proximal part of the femur. Located through the nail is a proximal lag screw, which locates through the nail and into the ball of the femur across the break. It allows dynamic compression (micromotion) of the fracture in a similar way to the compression hip screw (CHS) but uses an intramedullary nail instead of an external plate screwed on to the outside of the bone. The external plates require several screws to be inserted into strong cortical bone which further  weakens it. the intramedullary nail is self aligning during insertion as a result of the spongy central cancellous bone being much less strong and stiff than the cortical bone. The trochanteric femoral nail is designed for: Proximal femoral fractures, Femoral neck fractures, Pertrochanteric fractures, Intertrochanteric fractures, Subtrochanteric fractures, Ipsilateral fractures. These are typical in elderly patients as well as a result of trauma such as that resulting from motor cycle accidents in younger patients. (6) b) The sliding of the proximal lag screw through the nail allows dynamic compression of the fracture site encouraging bone growth. A second smaller diameter pin implanted parallel and proximally to the lag screw prevents rotation of the head-neck bone fragment. Intramedullary implant allowing minimally invasive approach using only 4/5 stab type entry points into the hip. This causes less disruption to soft tissue, less blood loss, shorter operating time, reduced risk of post-operative complications. Reduced distance between hip joint and mechanical axis of femur results in a lower bending moment which allows immediate mobilisation with full weight bearing Distal nail (lower down) allows flexibility-Reduced risk of intra and post-operative shaft fractures Short nail length -Reduced risk of intra and post-operative shaft fractures Small Distal nail allows insertion without reaming-Time saving technique and reduced blood loss Small Proximal nail diameter 13 mm which again means the operation is less invasive Universal design for both hips (no anteversion) resulting in a smaller inventory and a single straightforward surgical technique	6 marks 7 marks

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B3	Reamed/undreamed nail location allows the surgeon choice of approach Permanent device that does not have to be removed c. The intramedullary nail has the following high requirements: Fatigue resistance, high strength, good specific strength, moderate cost and similar stiffness to bone to facilitate load transfer. This is best achieved using a Titanium Alloy but the modulus mismatch is still high. The typical choice is Ti6Al4V, but other alloys can be considered. The part is forged from a bar stock to enhance the fatigue strength. Titanium has poor wear properties so it requires a TiMAX coating which enhances the products further. TiMAX is a hard oxide layer on the surface created by pickling in an alkali solution. This increases scratch resistance, fatigue resistance, makes the surface slide more easily. The formed final surface is machined after pickling. You would not cast nail as the fatigue properties would be much worse. The large diameter lag screw and the second smaller diameter lag screw and the distal screw are all machined from titanium typically a forged bar. Possibility of using stainless steels as well however you would end up with a galvanic cell with dissimilar metals.. The insertion jig is made from carbon fibre and PEEK composites. Future devices maybe made from resorbable materials like PLA or PGA. One final possibility is that they may be hollow metal tubes that can be inflated in place. Timing belts or timing chains are the part of an internal combustion engine that synchronises the rotation of the crankshaft and the camshaft so that the engine's valves open and close at the correct times during each cylinder's intake and exhaust strokes. a) Explain the advantages and disadvantages of using a timing belt in comparison with a timing chain. [6 marks] b) Describe using suitable sketch the construction used for the current generation of timing belts? In each part of the composite structure identify the choice of materials and give reasons as to why these materials are used. [10 marks] c) Future timing belts have to reduce their width and therefore the choice of materials that can be used is being revisited. Explain how the design is likely to develop over the next five years. [9 marks] a) Timing chains are prone to stretching. They can be extended by as much as 0.5% over about 100,000 km of usage. They exhibit high friction and therefore lose efficiency, they are also quite noisy in comparison with timing belts. They are however smaller in dimensions (in particular thickness) Timing belts are fatter, quieter, lighter, and more efficient due to lower frictional losses and as a consequence of their fibre reinforcement less prone to creep. They extend by less than 0.05% over about 100,000km usage. As a consequence the engine remains more fuel efficient. b) Current tech include E glass cords with ~6600 glass filaments, each of about 10μ diameter. These fibres are carefully bundled together and impregnated with rubber. The fibres are very stiff in tension but they are also soft in bending. They therefore provide significant reinforcement of rubber in the tensile direction, they prevent rubber creep and the latex rubber binds the cords together, prevents glass abrasion against neighbouring fibres and provides adhesion between cord and rubber. Historically the rubber was polychloroprene but in 	12 marks

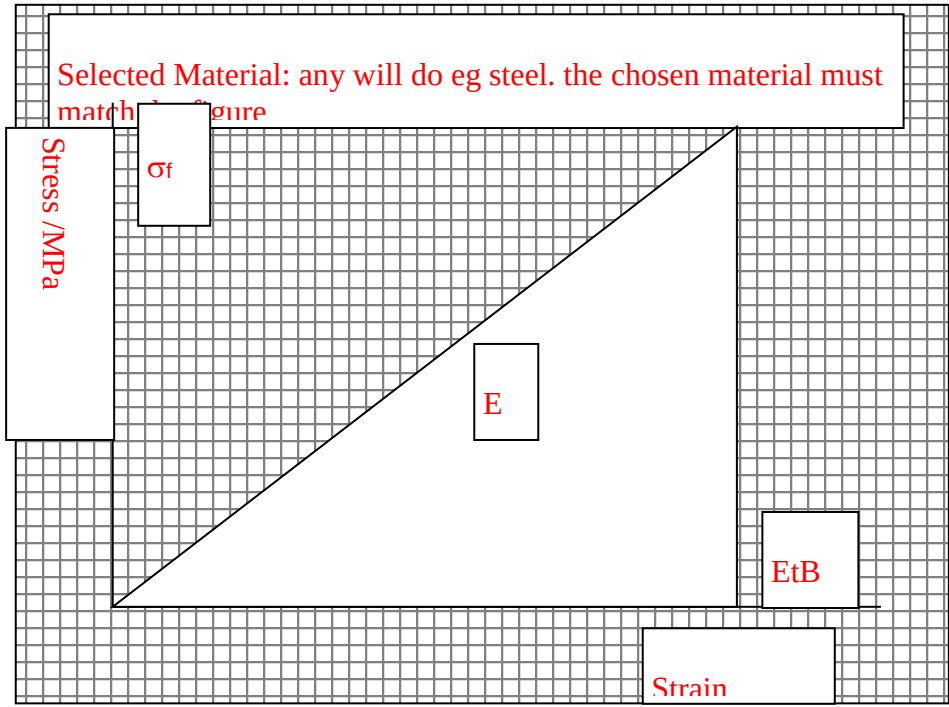


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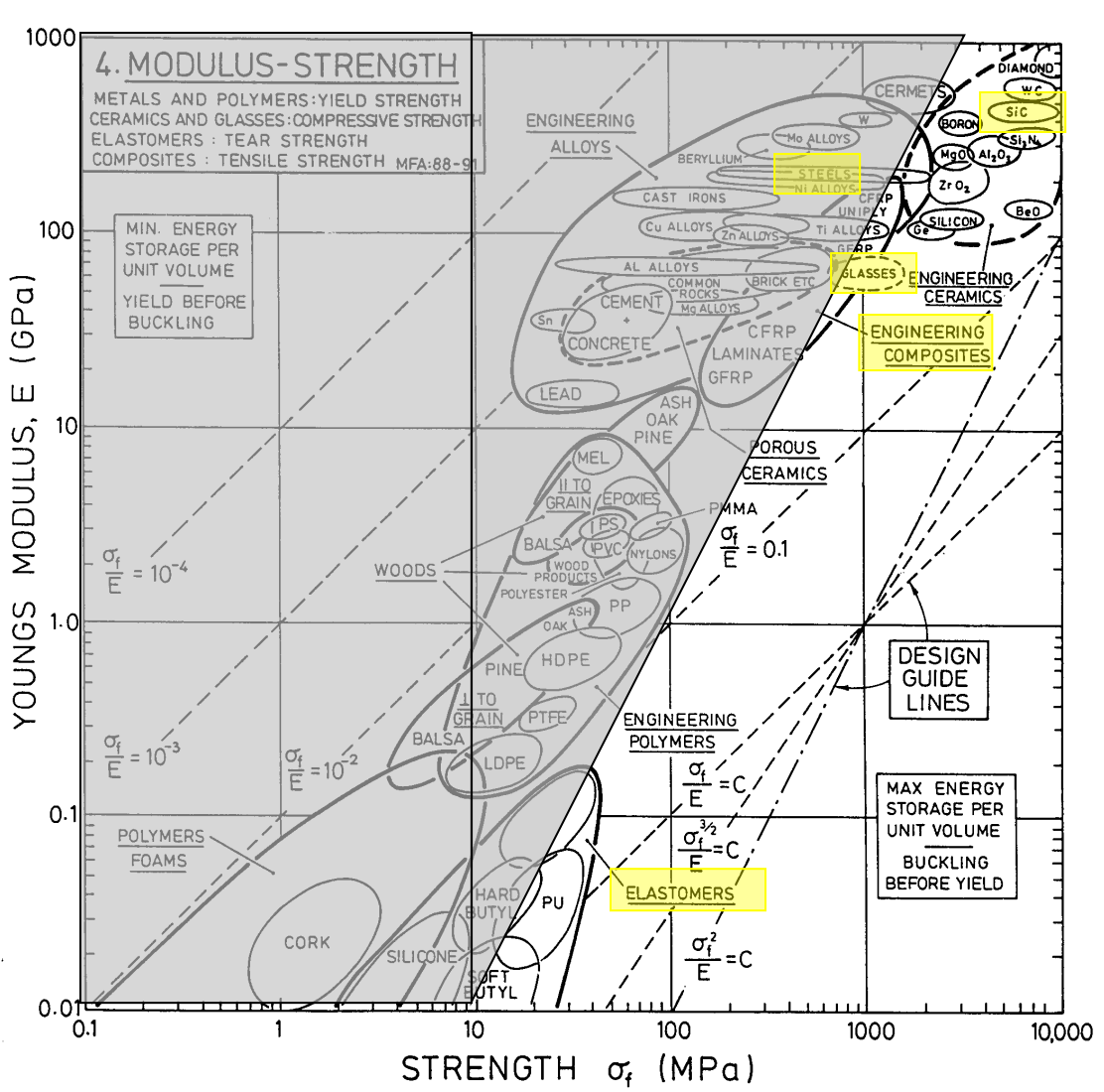
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	more recent times HNBR has been adopted as it exhibits good properties even when in contact with hot sump oil as It does not swell significantly in oil and has a good compatibility with high temperatures and the complex chemical environment of the engine. The HNBR is not fully saturated so that it can continue to be crosslinked with sulphur bridges. More recently the crosslinking has become peroxide based and the latex is moving to HNBR latex as well. The fabric has also evolved from PU and PTFE hybrid designs to more conventional polyester or nylon designs c) Next generation will almost certainly replace the glass fibres with even stiffer and more creep resistant cord materials like carbon fibre. The main issue there is that the fibres are stiffer but carbon has a lower strain to break. One potential issue is therefore that the fixed amount of bend in the belt system might result in fatigue in the carbon fibres within the cord. This is an ongoing investigation. The second solution would be to have hybrid cord bundles with carbon in the core and glass around the perimeter. These might be better in the bending than pure carbon cords. Other improvements include the actual tooth design to reduce fatigue stretching of the teeth, different latex in the cords (move to an HNBR material) to provide better bonding to the rubber material and finally revised fabric constructions	



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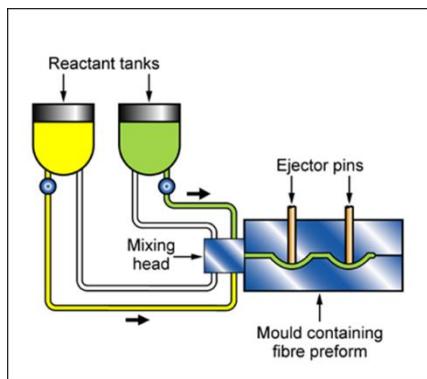
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C1	Question 1. A spring is a device that stores elastic energy. Sketch a suitable linear elastic stress versus strain graph for any specified material and identify the maximum elastic energy density, W that can be stored in the spring.  Derive below the following equation from your figure for the elastically stored energy expressed only in terms of the modulus, E, and the failure strength, σ_f, of the material. $W = 0.5 \sigma_f^2 / E$ Energy stored = W = Area of a triangle = $0.5 \text{ base} \times \text{height} = 0.5 \text{ EtB} \times \sigma_f$ $E = \sigma_f / \text{EtB}$ Therefore $W = \frac{1}{2} \frac{\sigma_f^2}{E}$. (2 marks) Using the above relationship identify a suitable materials selection parameter for a spring that has a minimum volume design which can store the maximum energy. M1 minimum σ_f^2 / E (1 mark)	2
Question C2	Question 2. Using the Ashby diagram given, identify the most suitable materials from a range of materials classes that can be used as a minimum volume spring material. You	1

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Question C3.	should indicate on the chart using lines to define regions of interest, what you might think are the optimum materials and you should highlight your chosen materials. 	2 for line, shading and picking best options								
	<table border="1"> <thead> <tr> <th>Selected Material</th><th>Index/MPa</th><th>Comments on suitability</th></tr> </thead> <tbody> <tr> <td>Rubbers</td><td>20</td><td>Good materials but includes some damping, used often in vibration control.</td></tr> <tr> <td>Glasses</td><td>10</td><td>Weak in tension and bending, with very low losses and so used in galvanometers.</td></tr> </tbody> </table>	Selected Material	Index/MPa	Comments on suitability	Rubbers	20	Good materials but includes some damping, used often in vibration control.	Glasses	10	Weak in tension and bending, with very low losses and so used in galvanometers.
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	Carbon Fibre RP	8	Comparable to steel but more expensive.	4 marks in total or 1 per material
	Nylon	3	Low cost and easy to make, used in toys for example.	
	BeO & other eng ceramics	10-100	All weak in tension	
	Glass Fibre RP	5	Similar to CFRP but much cheaper.	
	Spring Steel	10	Very widely used from suspensions in building, trains and planes to watches.	
	Titanium	10	Excellent prestige use material, with good corrosion properties.	
Question 4				
a) Identify the most suitable processing technique that should be used to make a composite door panel for a lightweight sports car such as a Lotus.				
Resin transfer moulding				
b) Sketch below your chosen technique highlighting any key features.				
1 mark				
 c) Briefly describe the key features of this process The dry reinforcement and any inserts are added to a mould made from light alloy or a glass reinforced plastic The carefully mixed resins are combined and pumped under a low pressure (2MPa) into the mould. It must have a low viscosity. The press provides a clamping pressure during mould filling.				
Usually done at room temperature but cycle times can be reduced by heating the tools.				
Ensuring that fibres retain their position in a tool and to allow rapid positioning of fibres has led to the increasing use of preformed fibre mats.				
3 marks				



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	Question C5. a) Identify the most suitable processing technique that should be used to make carbonated drinks can out of a sheet of rolled aluminium. deep drawing b) Sketch below your chosen technique and briefly describe the key features of this process. c) Briefly describe the key features of this process. In DEEP DRAWING, a flat sheet-metal blank is converted into a cylindrical or box-shaped part by drawing it through a die with a tool-steel punch. The process relies on the plastic stretching of the blank, which is limited by the limiting draw ratio (LDR) of the material of the blank. The process is known for producing aluminum beverage cans. Further drawing operations include 'redrawing' to obtain greater depth than permitted by the LDR without changing the wall thickness, and 'ironing' in which the thickness of the sheet is reduced thus resulting in a deeper cup.	1 mark 3 marks 3 marks